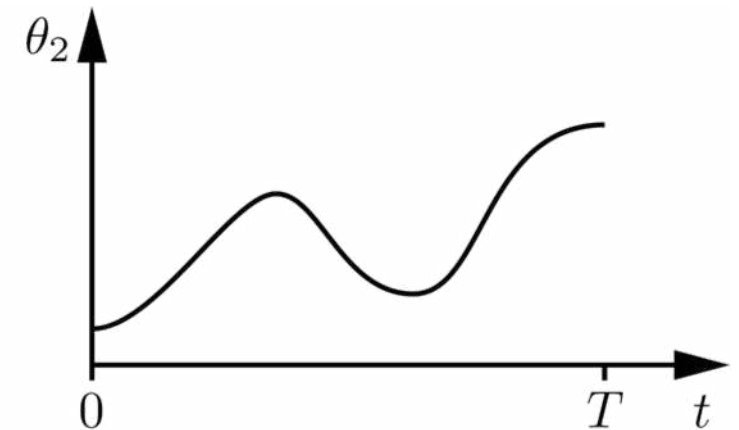
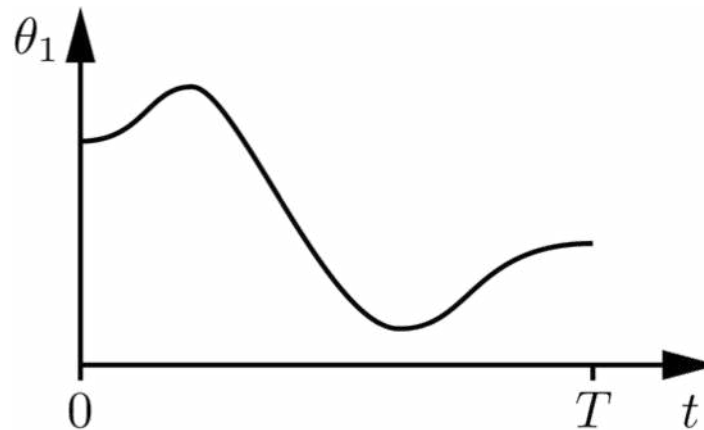
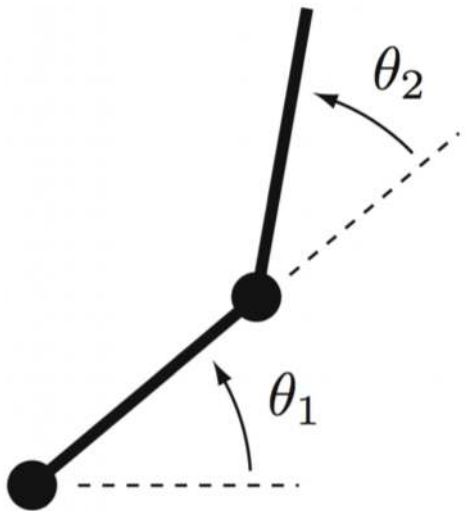


Chapter 2	Configuration Space
Chapter 3	Rigid-Body Motions
Chapter 4	Forward Kinematics
Chapter 5	Velocity Kinematics and Statics
Chapter 6	Inverse Kinematics
Chapter 7	Kinematics of Closed Chains
Chapter 8	Dynamics of Open Chains
Chapter 9	Trajectory Generation
	9.1 Definitions
	9.2 Point-to-Point Trajectories
	9.3 Polynomial Via Point Trajectories
Chapter 10	Motion Planning
Chapter 11	Robot Control
Chapter 12	Grasping and Manipulation
Chapter 13	Wheeled Mobile Robots

Important concepts, symbols, and equations

Trajectory: A specification of the configuration as a function of time.

$$\theta(t), t \in [0, T]$$

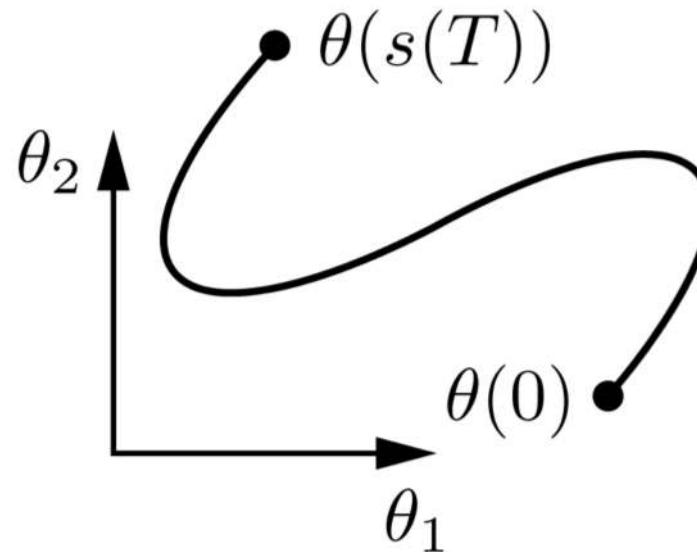
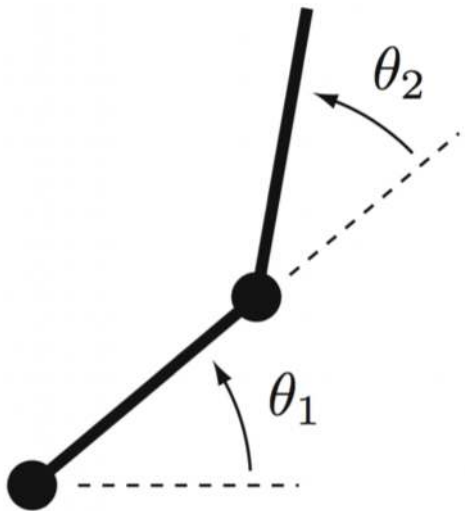


Important concepts, symbols, and equations (cont.)

Path: A specification of the configuration as a function of a path parameter.

$$\theta(s), \quad s \in [0, 1]$$

Time scaling: A mapping $s(t): [0, T] \rightarrow [0, 1]$, from time to the path parameter.



Important concepts, symbols, and equations (cont.)

Motion as a function of $\theta(s)$ and $s(t)$:

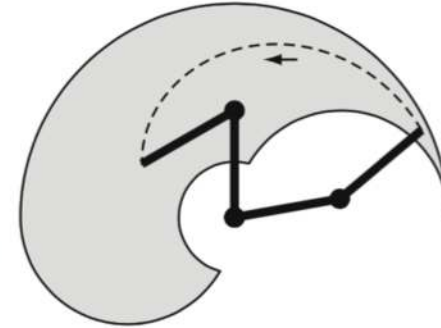
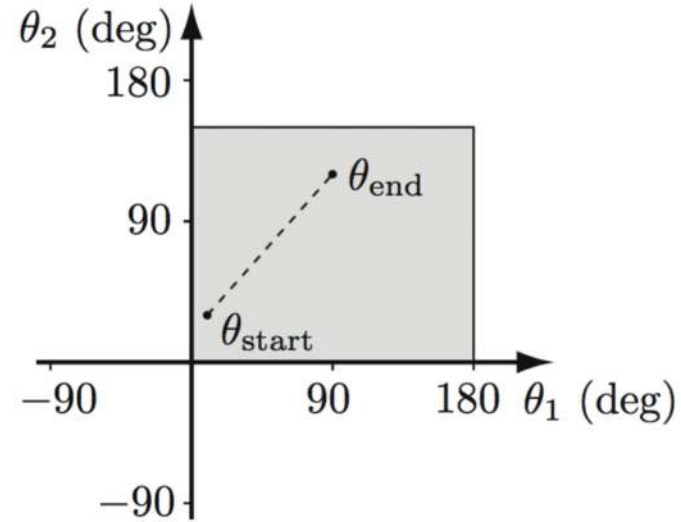
$$\dot{\theta} = \frac{d\theta}{ds} \dot{s},$$
$$\ddot{\theta} = \frac{d\theta}{ds} \ddot{s} + \frac{d^2\theta}{ds^2} \dot{s}^2$$

Both $\theta(s)$ and $s(t)$ must be twice-differentiable.

Important concepts, symbols, and equations (cont.)

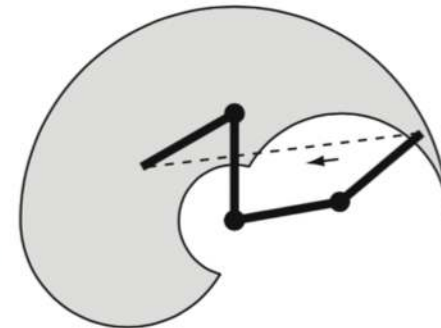
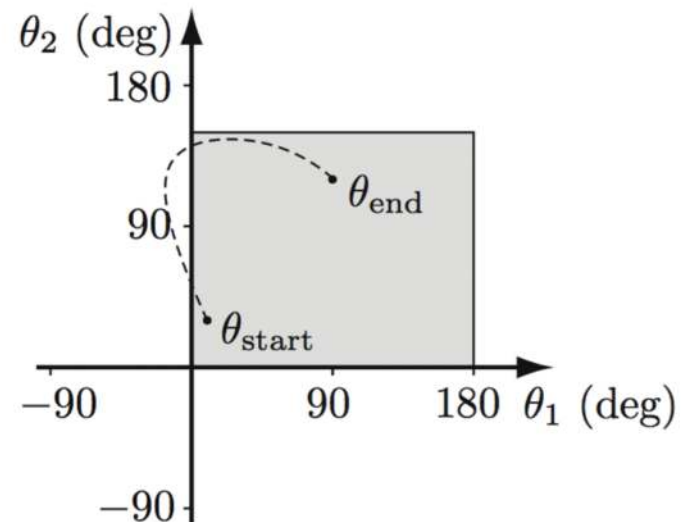
straight line in joint space

$$\theta(s) = \theta_{\text{start}} + s(\theta_{\text{end}} - \theta_{\text{start}})$$

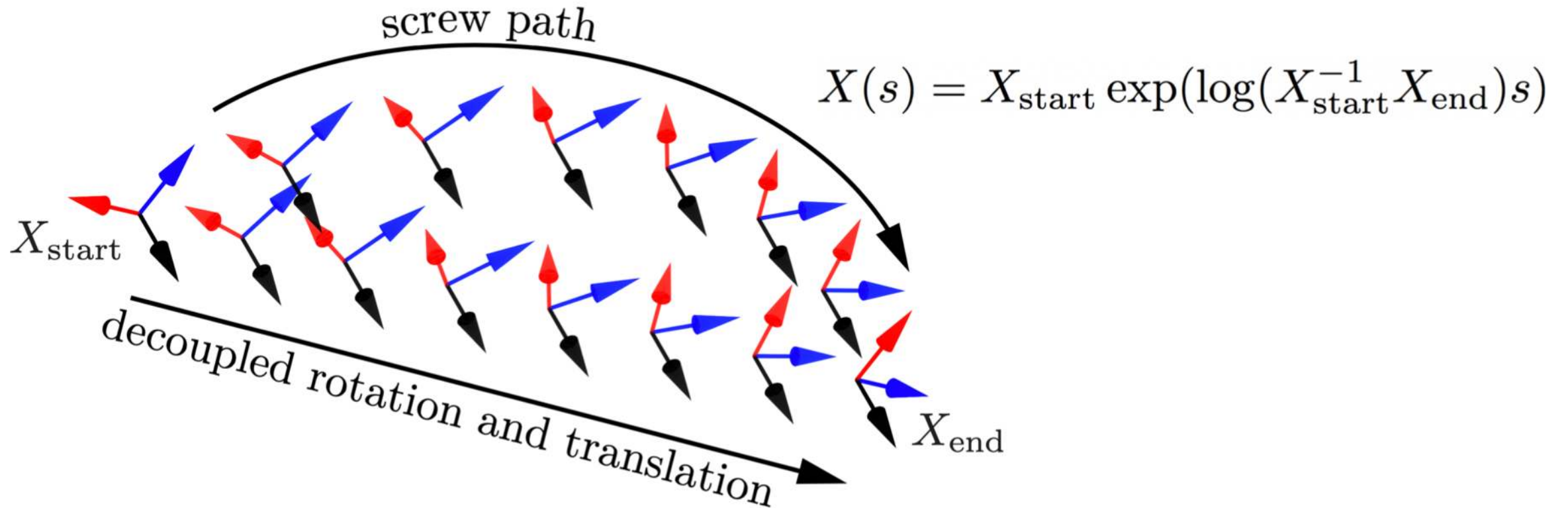


straight line in task space

$$X(s) = X_{\text{start}} + s(X_{\text{end}} - X_{\text{start}})$$



Important concepts, symbols, and equations (cont.)



$$p(s) = p_{\text{start}} + s(p_{\text{end}} - p_{\text{start}}),$$

$$R(s) = R_{\text{start}} \exp(\log(R_{\text{start}}^T R_{\text{end}})s)$$

Important concepts, symbols, and equations (cont.)

Third-order polynomial time scaling

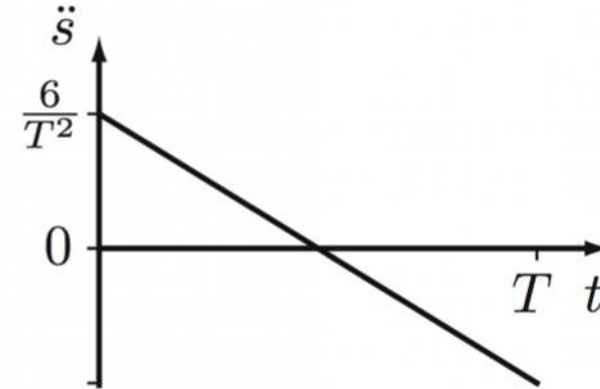
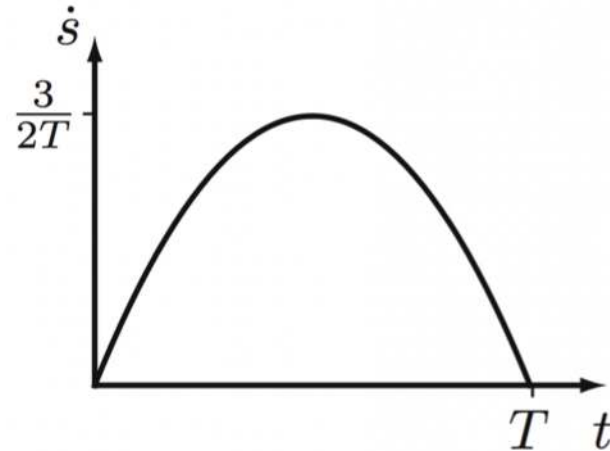
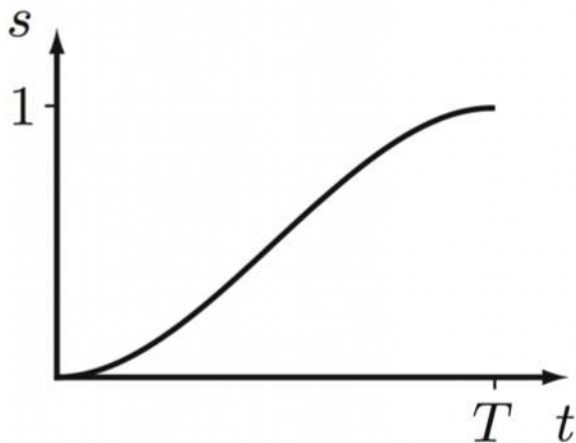
$$s(t) = a_0 + a_1t + a_2t^2 + a_3t^3$$

$$\dot{s}(t) = a_1 + 2a_2t + 3a_3t^2$$

$$a_0 = 0, \quad a_1 = 0, \quad a_2 = \frac{3}{T^2}, \quad a_3 = -\frac{2}{T^3}$$

$$s(0) = 0 \quad \dot{s}(0) = 0$$

$$s(T) = 1 \quad \dot{s}(T) = 0$$

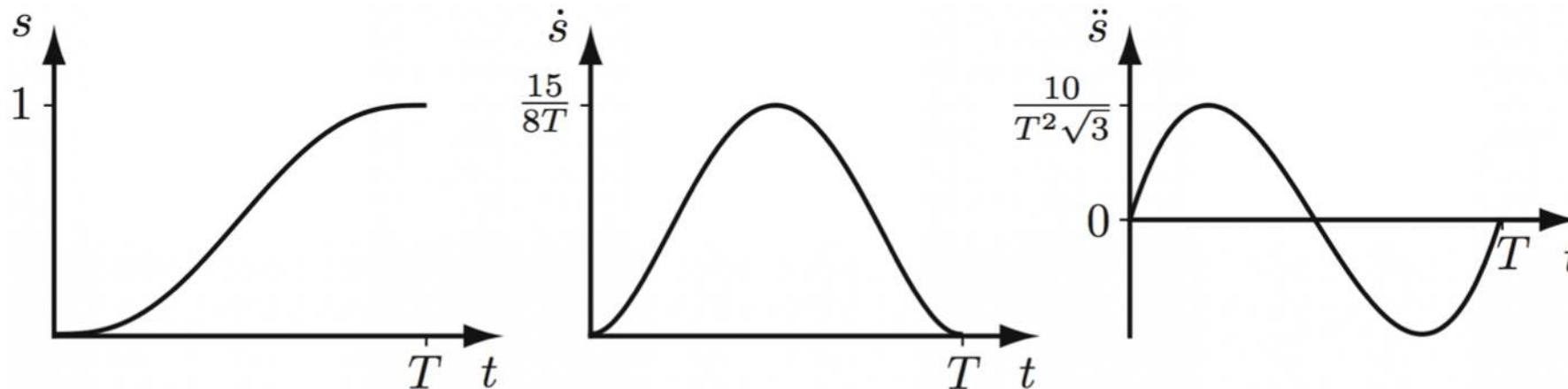


Important concepts, symbols, and equations (cont.)

Fifth-order polynomial time scaling

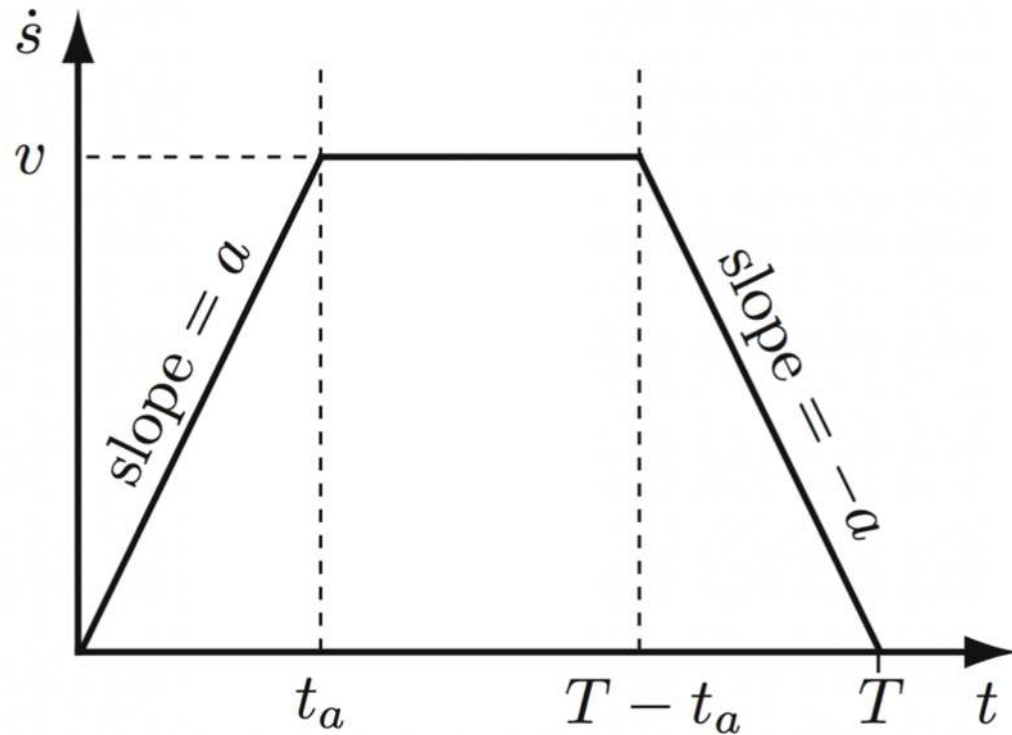
$$s(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5$$

$$\begin{aligned} s(0) &= 0 & \dot{s}(0) &= 0 & \ddot{s}(0) &= 0 \\ s(T) &= 1 & \dot{s}(T) &= 0 & \ddot{s}(T) &= 0 \end{aligned}$$



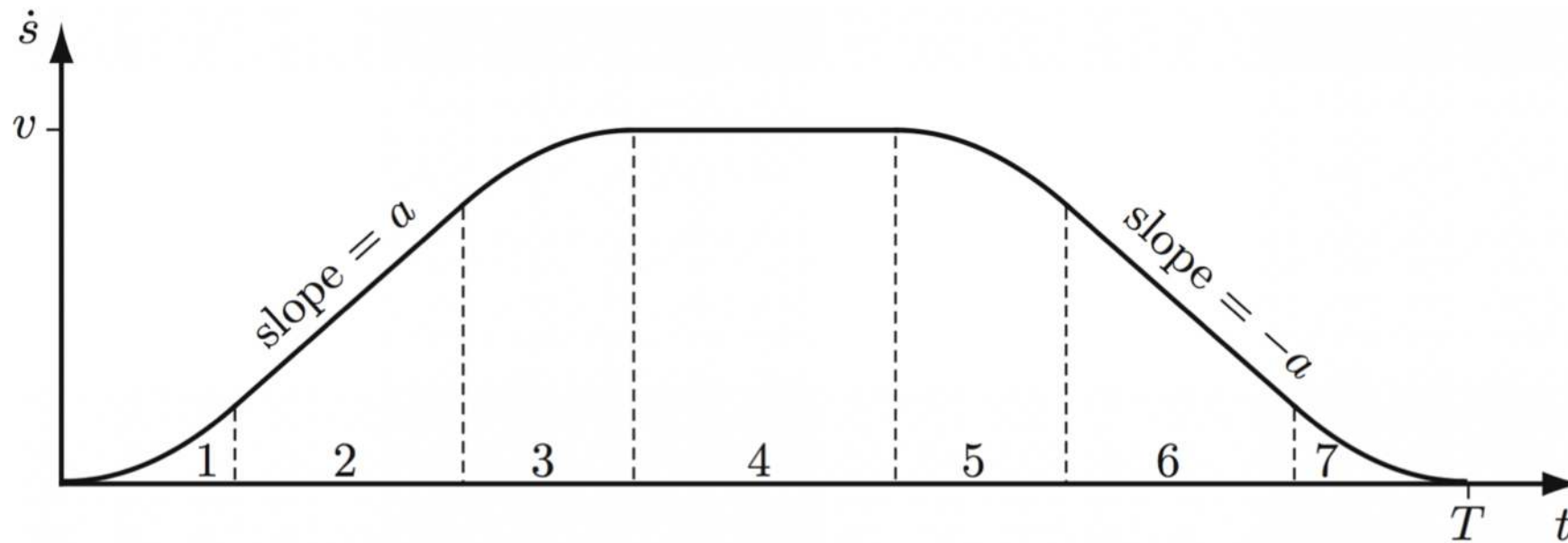
Important concepts, symbols, and equations (cont.)

Trapezoidal time scaling



Important concepts, symbols, and equations (cont.)

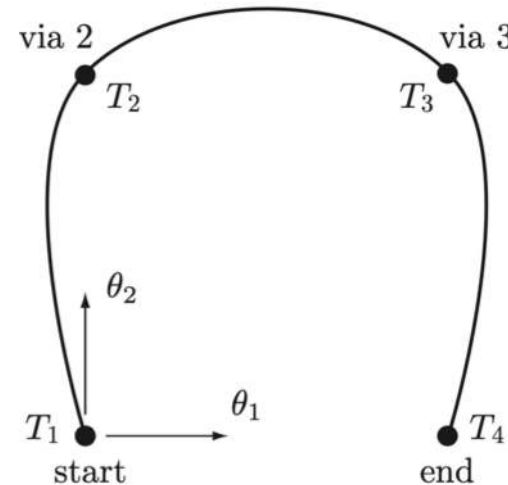
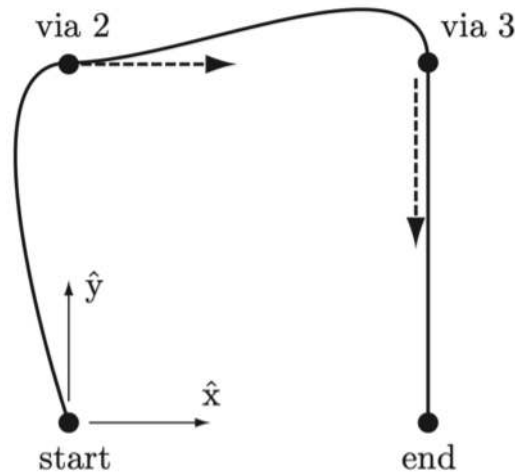
S-curve time scaling



Important concepts, symbols, and equations (cont.)

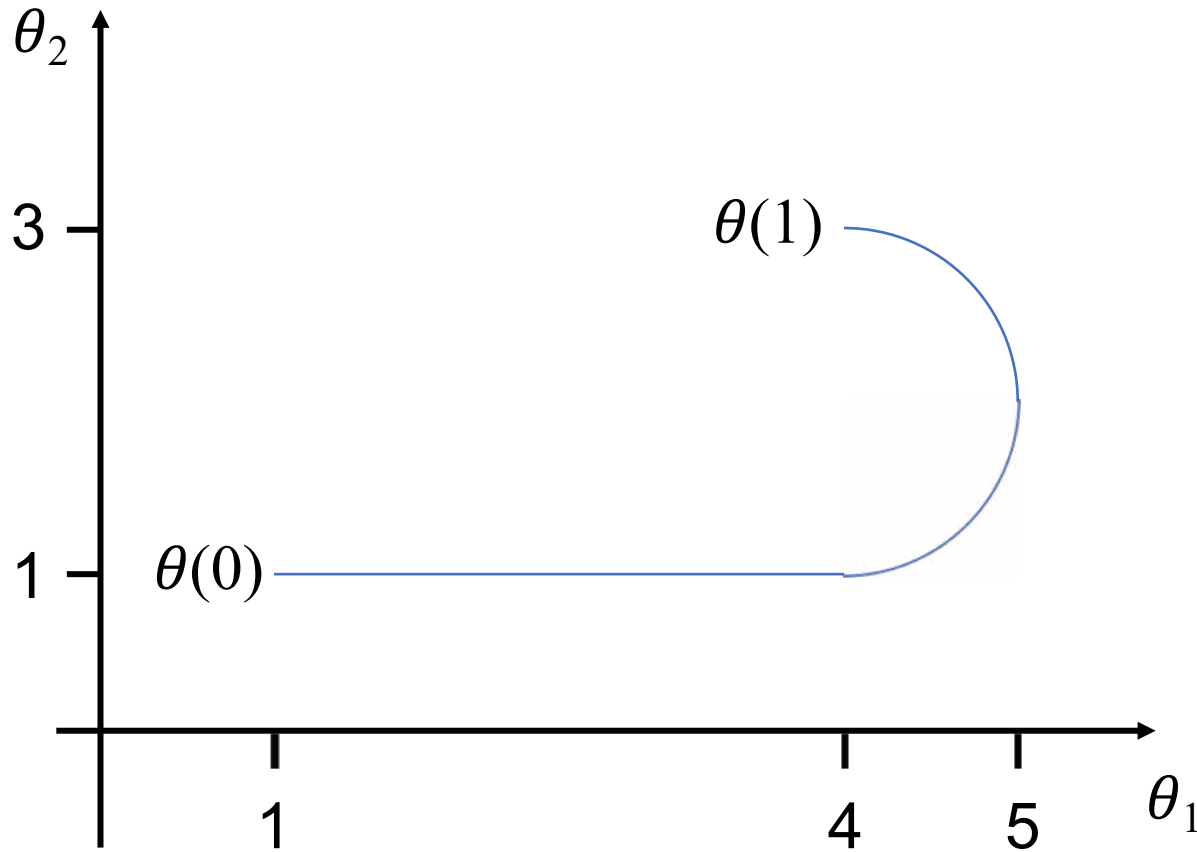
Polynomial interpolation through via points

- third-order interpolation using via times, configurations, and velocities
- third-order interpolation using via times, configurations, and equal velocities and accelerations before and after vias



Many other methods, including **B-splines** (paths stay within convex hull of control points, but don't pass through them).

Give an expression for the path $\theta(s)$, $s \in [0,1]$.



What kind of time scaling can be used to obtain a continuous jerk profile?

What is the maximum joint velocity obtained on a straight-line rest-to-rest trajectory with cubic polynomial time scaling?

Describe a circumstance under which the coast phase of the trapezoidal time scaling is not used.

Give an equation to implement a third-order polynomial time-scaled rest-to-rest motion following a screw axis.

A time scaling can be written as $s(t)$ or $\dot{s}(s)$. If $s(t) = at^2$, what is $\dot{s}(s)$?